Progress Report

Sensor Trust for Lunar South Pole Landing - state of the LANTERN testbed on 18 October 2026

Summary

The Unity side is working and verified: a lander descends from 102.5 km above a flat spot on the Site04 crater floor and touches down at 0.50 m/s under autopilot, with a live-scan position fix (LUMEN) and per-run data logging. A fast Python test bench for the 500-run comparison is written and was smoke-tested on 40-100 runs, but **the full Monte Carlo evaluation, the ML classifier, the tuning, the sweeps and the Unity replay have not been run**. No result in this report from the Python bench should be quoted as a finding.

1. Verified in Unity

Item	Result	How it was checked
Unity connection and Play mode	Works. Play did not advance while the editor window was inactive; Application.runInBackground fixes it and is now set by the autopilot	Frame counter and Time.time sampled in Play
Landing site	Unity (6198, 8512), 1.88 km from the tile centre, 1729.5 m elevation, mean slope 1.1 deg, 45 m-window maximum 2.8 deg. The tile centre itself is 33 deg	Slope derived from the DEM (the supplied slope file is truncated); Unity terrain height at the point equals the DEM
Shadows	Low-Sun cast shadows render across the terrain	Camera captures
Autopilot, 500 m start	First run touched down at 2.07 m/s (over target); after retuning, 0.50 m/s	Console and sampled state
Autopilot, 102.5 km start	Touchdown at t = 583-587 s, 0.50 m/s vertical, 0 m/s horizontal, 0.0 deg tilt, 251 kg fuel left (549 kg used), exactly on the target. Coast about 190 s to 312 m/s, then about 0.69 throttle to touchdown	Two full runs and the logged CSV
LUMEN live scan	Shift opposed the injected error (+60, -40) m: (-64, 32) m at 390 m and 272 m altitude, confidence 1.00. After the altitude-scaled cell fix it gets a fix from 2463 m down; error 47 m at 2205 m (41 m cell), 14.5 m at 921 m, 4 m at 155 m (8 m cell). Valid in 9% of logged rows; median error 8.9 m; max 48.7 m	Logged run CSV (108 valid fixes)
HUD	Text now scales to 3% of screen height with a dark backing box and is readable at 1920x1080	Game-view screenshot (out/hud_check.png)
Data logging	TelemetryLogger writes per-run CSV and runs_summary.csv; flushed per row. The first run's summary line recorded a start altitude of 0 (logging bug, fixed afterwards, fix not re-verified)	Files in out/runs/

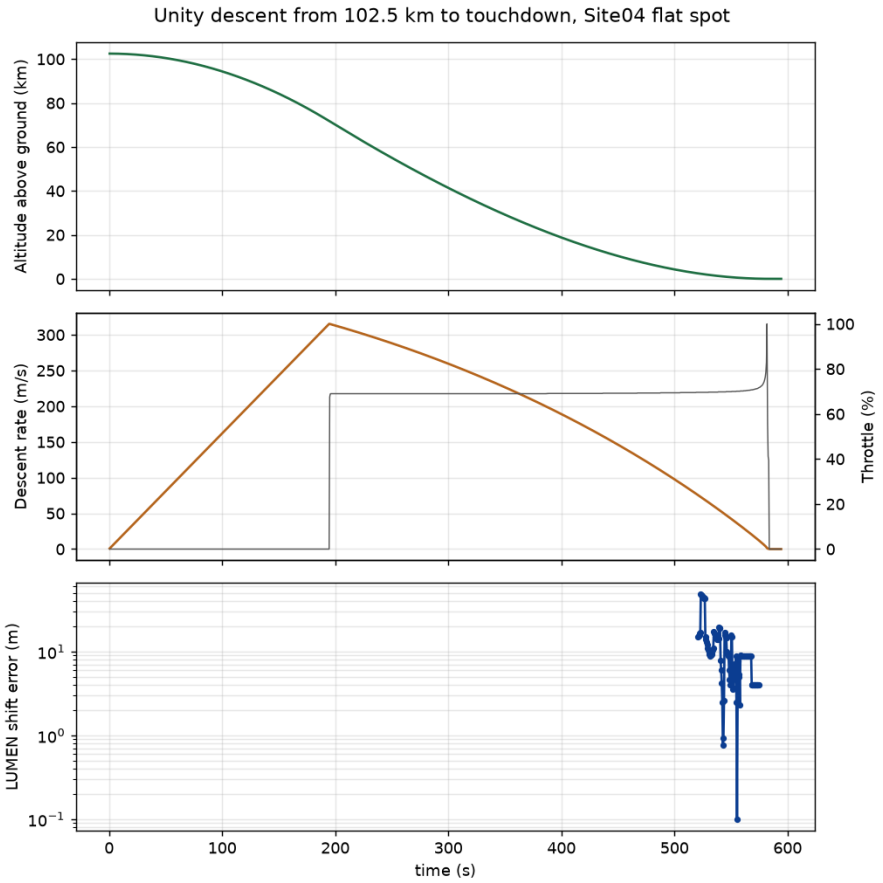


Figure 2. Logged Unity descent from 102.5 km: altitude, descent rate with throttle, and LUMEN position error.

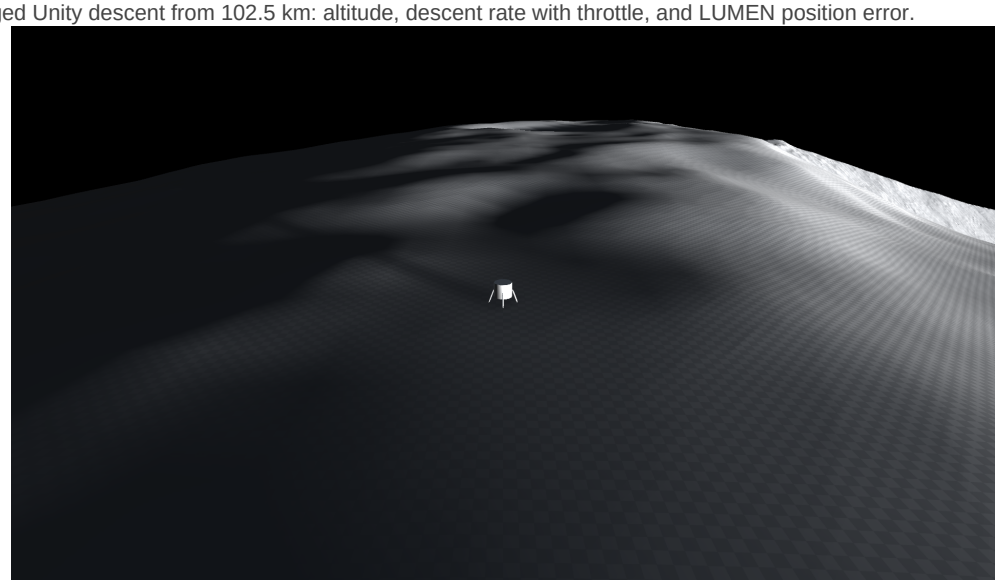


Figure 3. Lander on the Site04 crater-floor landing spot (Unity camera capture).

2. Built but not run at scale

Python test bench (sim/). world.py (Site04 and procedural crater worlds, onboard map with registration shift and noise, unmapped craters, coarse reference surface, shadow and lighting statistics), engine.py (batched truth physics, IMU, altimeter, camera and live-scan sensors, fault bursts, delay, dropout, a 9-state EKF, guidance on the estimate, seven judges plus three ablations), pipeline.py (parallel ML, tuning, main, sweeps and replay stages) and plots.py. Scenarios: cliff crossing (HAKUTO-R), cliff with frequent faults, shadow with noisy altimeter (IM-2), delayed readings (ispace M2), altimeter dropout (IM-1).

Unity replay. ReplayPlayer.cs and ReplaySceneBuilder.cs compile without errors; the replay scene has not been built because no replay data has been exported yet.

Smoke test only (40-100 runs, Site04, Sun from the west, not tuned, not a result). The fixed-gate baseline rejected about 99% of valid readings and crashed in nearly every run, which matches the HAKUTO-R failure mechanism. The trust layer landed more often than the PITFAL-style filter in the cliff and shadow scenarios, but scored 0% on delayed readings and about half on dropout. These early numbers came from a model whose guidance and sensor settings were chosen while looking at Site04 runs, so they need the tuning-on-procedural-only procedure before any claim.

3. Not done

- The 500-run Monte Carlo on Site04, ablation table, lighting sweep, position-uncertainty sweep, wrong/missing-map and divert tests.
- ML classifier training (a run was started and stopped at the user's request; no model was produced) and tuning on procedural worlds.
- Site01 tuning data (the tile is not in the repository); the HAKUTO-R altitude-curve replay (debriefing data not available).
- Building and running the Unity replay scene; the in-flight HUD showing true vs estimated altitude, gate width and per-sensor trust bars.
- LUMEN has no fix below roughly 80 m (footprint smaller than the 40 cells needed) and none above about 2.5 km (by design); only partly investigated.

4. Deviations and caveats

- The plan scopes powered descent from about 15 km; the Unity demonstration starts at 102.5 km at the user's request, from rest, falling straight down. A real descent from 100 km altitude starts with about 1.7 km/s of horizontal speed, which the autopilot does not handle. The Python bench starts at 15 km.
- Several Unity checks (HUD, camera) rely on one screenshot each; the camera capture tool did not appear to follow the lander during Play.
- The simulated 'real Moon' differs from the onboard map only by synthetic craters, noise and shifts, not by real unmapped terrain.

5. Next steps

- Run ML training and tuning on procedural worlds, then the Site04 evaluation with fixed seeds.
- Export a replay where the baseline crashes and the trust layer lands; build and capture the Unity replay scene.
- Write up results with confidence intervals and the stated limits.

6. Repository map

Path	Contents
LANTERN/	Unity 6 project (scene Site04, scripts, editor builders)
sim/	Python test bench and experiment pipeline
tools/	Heightmap and rendering scripts, PDF generator
deliverables/	Project plan PDFs (literature review, gap, objectives, methodology, summaries)
docs/	This project description and progress report
out/	Plots, telemetry runs, figures (large caches excluded from git)
data/	NASA PGDA GeoTIFFs (excluded from git; see README)